A Software Approach to the PPT2 Conjecture

Noah Novšak

Master's study programme Computer and Information Science, study field Data Science

Faculty of Computer and Information Science

Supervisor: asist. prof. dr. Aljaž Zalar
Co-supervisor: prof. dr. Igor Klep

The PPT2 conjecture asserts that the composition of any two PPT maps is entanglement breaking. It is proven for maps on matrices of size up to 3 × 3 and for several structured families, but the general case remains open; the smallest open case, 4 × 4, is the one this thesis attacks computationally. We build a reproducible Julia pipeline that (i) mass-produces provably indecomposable entanglement witnesses via the Klep–McCullough–Šivic–Zalar construction of positive but not completely positive maps, rationalizing each certificate after the semidefinite program is solved so that every stored witness is exact – 10,000 witnesses in under an hour, orders of magnitude faster than comparable implementations; (ii) generates bound entangled PPT candidates by generic random sampling, partial-transpose-invariant sampling, and witness-guided extraction; and (iii) tests the conjecture both by screening tens of thousands of composed channels with witness and DPS criteria, and by a see-saw SDP that searches the manifold of composed PPT maps directly. No counterexample is found. The witness library proves to be a collection of single-state detectors: each witness detects essentially only the state extracted from it. The central finding is a sharp contrast: every one of the 10,000 witnesses attains a negative optimum somewhere on the PPT cone, yet not one fires on the composition manifold – precisely the signature expected if the conjecture holds in dimension four. This indicates that a counterexample, if one exists at all, must inhabit a measure-zero subset of the composition manifold that a random search cannot reach. As a by-product, the construction yields explicit biquadratic 4 × 4-forms that are positive but not sums of squares, which is a notoriously difficult problem in real algebraic geometry.

PPT2 conjecture | quantum entanglement | positive maps | semidefinite programming | bound entanglement

Programski pristop k domnevi PPT2

Domneva PPT2 trdi, da kompozitum poljubnih dveh PPT-preslikav, uniči prepletenost. Dokazana je za preslikave na matrikah velikosti do 3 × 3 in za več strukturiranih družin, v splošnem pa ostaja odprta; najmanjši odprti primer, 4 × 4, v tem delu napademo računsko. Razvijemo ponovljiv cevovod v jeziku Julia, ki (i) s konstrukcijo Klepa, McCullougha, Šivica in Zalarja množično izdeluje dokazano nerazcepne priče prepletenosti iz pozitivnih, a ne popolnoma pozitivnih preslikav; vsak certifikat po rešitvi semidefinitnega programa racionaliziramo, tako da je vsaka shranjena priča eksaktna – 10.000 prič zgradimo v manj kot uri, za rede velikosti hitreje od primerljivih implementacij; (ii) generira mejno prepletene PPT-kandidate z generičnim naključnim vzorčenjem, z vzorčenjem, invariantnim na delno transpozicijo, ter z ekstrakcijo iz prič; in (iii) domnevo preizkusi s presejanjem več deset tisoč kompozitumov ter z izmenično optimizacijo, ki mnogoterost kompozitumov PPT-preslikav preiskuje neposredno. Protiprimera ne najdemo. Knjižnica prič se izkaže za zbirko detektorjev enega samega stanja: vsaka priča zazna v bistvu le stanje, ki je bilo iz nje ekstrahirano. Osrednja ugotovitev je ostro nasprotje: vsaka od 10.000 prič doseže negativni optimum nekje na stožcu PPT, na mnogoterosti kompozitumov pa se ne sproži nobena – natanko kar pričakujemo, če domneva v dimenziji štiri drži. To nakazuje, da morebitni protiprimer, če sploh obstaja, leži na podmnožici kompozitumov z mero nič, ki je naključno iskanje ne doseže. Spotoma konstrukcija ustvari eksplicitne bikvadratne 4 × 4-forme, ki so pozitivne, a niso vsote kvadratov, kar je znano težek problem v realni algebraični geometriji.

domneva PPT2 | kvantna prepletenost | pozitivne preslikave | semidefinitno programiranje | mejna prepletenost

Quantum entanglement is the central nonclassical resource of quantum information theory: it underlies teleportation, dense coding, and quantum cryptography, and it provides a structural lens for understanding quantum channels [1]. Whether a given bipartite state is entangled at all – the *separability problem* – is therefore a foundational question, and a computationally formidable one: deciding separability is NP-hard [2, 3]. The workhorse necessary condition is the positive-partial-transpose (PPT) criterion [4], which is also sufficient in dimensions 2×2 and 2×3 [5], but not beyond: there exist entangled states whose partial transpose stays positive. No pure entanglement can be distilled from such *bound entangled* states [6], yet they are far from inert – they can fuel secret-key distillation [7], private states and data hiding [8], and quantum metrology [9].

This thesis studies the PPT2 conjecture, posed by Christandl in 2012 [10]: whether the composition of two PPT maps is always entanglement breaking [11]. The conjecture has a clear operational interpretation, since entanglement-breaking maps destroy all bipartite entanglement with any reference system; it asks, in effect, whether the weak, bound entanglement that a PPT map can transmit ever survives a second one.

The stakes are structural. Were the conjecture to hold, composing two PPT maps would always destroy entanglement: the PPT cone, though strictly larger than the set of entanglement-breaking maps, would collapse into it after a single self-composition – a strong rigidity statement about how far a PPT map can sit from being entanglement breaking. Were it to fail, the counterexample would be a pair of PPT maps whose composition is itself PPT yet entangled, a bound entangled channel assembled from ordinary PPT ingredients, bearing on the structure of quantum channels and on the distillability questions that turn on the PPT property [1, 6]. Either way the answer sharpens the picture of the boundary between positive and completely positive maps.

A decade of partial results maps out where the conjecture is settled. It holds for maps on M_2, where PPT and separability coincide [5], and on M_3, proven independently in [11] and [12]. In all dimensions it holds for structured families: Choi-type maps [13] (a family built on diagonal unitary symmetry [14]), Gaussian channels [11], and Cartan-covariant channels [15]. Asymptotic and generic versions are also known: the iterates of any unital or trace-preserving PPT map approach the entanglement-breaking set [16], every unital PPT channel becomes entanglement breaking after finitely many compositions [17], and compositions of independent random PPT states are generically separable [18]. The general case remains open for $n \geq 4$, where the geometry of positive maps grows markedly more intricate and direct analytic classification has so far resisted. The smallest open case, 4×4, is the natural target for a computational attack, and the one we pursue.

The computational machinery for such an attack exists, scattered across two literatures. On the quantum-information side, separability admits a complete hierarchy of semidefinite-programming (SDP) relaxations, the DPS hierarchy [19, 20], and every entanglement witness corresponds, through the Choi-Jamiołkowski isomorphism, to a positive but not completely positive (PNCP) map [1]. On the real-algebraic side, Klep, McCullough, Šivic, and Zalar [21] turned that correspondence into an algorithm: a construction that mass-produces PNCP maps from non-negative polynomials that are not sums of squares, each map an entanglement witness. A MATLAB prototype of the construction exists [22], and a contemporaneous, independent implementation has recently been used for entanglement detection [23] – though not, to our knowledge, for a systematic PPT2 search. Direct computational probes of the conjecture itself have so far been limited to random sampling in high dimensions [24].

The core challenge is twofold. First, separability testing is hard precisely in the regimes where the conjecture remains open, so every practical test is a one-sided relaxation. Second, numerical certificates near feasibility boundaries are fragile. Therefore, this thesis focuses on methodology: how to construct, compose, test, and validate candidate objects in a reproducible way.

The main contributions are:

1. A reproducible Julia workflow covering the whole search: witness generation, candidate generation, map composition, and entanglement checks, with exact post-solver rationalization of every certificate. The witness library it produces – 10,000 provably indecomposable witnesses in under an hour – is orders of magnitude faster to build than comparable implementations of the same construction [22, 23].
2. A direct comparison of three sources of bound entangled PPT candidates (generic sampling, partial-transpose-invariant sampling, and witness-guided extraction) and of three entanglement detectors (the scalar witness test, the map witness test, and the DPS relaxation [20]), quantifying the cost and reach of each.
3. A two-pronged test of the conjecture in the smallest open dimension: an all-pairs composition scan over the candidate pools, and a see-saw SDP that searches the manifold of composed PPT maps directly. Neither finds a counterexample, and the see-saw exposes a sharp, previously unreported contrast: witnesses that fire everywhere on the PPT cone go uniformly silent on the composition manifold, indicating that a counterexample, if one exists, must lie on a measure-zero subset beyond the reach of a random search.
4. A bulk supply of explicit biquadratic $(4, 4)$-forms that are positive but not sums of squares. Non-negative polynomials that are not SOS are hard to exhibit, and the construction produces them generically and in quantity in the 4×4 case.

The rest of the thesis is organized as follows. Section 1 develops the mathematical background: notation, matrix spaces, linear maps, entanglement, the criteria used to detect it, and the semidefinite-programming machinery, including the polynomial route to witnesses. Section 2 presents the computational methods and the implementation. Section 3 reports the experimental results and discusses their significance, the limitations, and directions for future work.

1. Theoretical Background

This section establishes the mathematical framework underpinning the PPT2 conjecture and its computational study. We fix notation and define the key objects: linear maps, their positivity properties, quantum states, and entanglement criteria, then develop the semidefinite programming tools used in the implementation.

1.1. Preliminaries. We write $\mathbb{C}^n$ and $\mathbb{R}^n$ for the complex and real coordinate spaces. $M_n(\mathbb{F})$ is the algebra of $n \times n$ matrices over the field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, abbreviated M_n when the field is clear from context. A linear map between matrix algebras is written $\Phi : M_n \to M_m$. We write $E_{ij} \in M_n$ for the matrix unit with a 1 in position (i, j) and zeros elsewhere, and $I_n \in M_n$ for the identity matrix, dropping the subscript to I when the dimension is clear from context.

In some parts of this work we also use *Dirac notation*, the standard convention in quantum mechanics. Let $|\psi\rangle$ denote the column vector $\psi \in \mathbb{C}^n$ and $\langle\psi|$ its adjoint row vector ψ^*. Then $\langle\psi \mid \varphi\rangle = \psi^*\varphi \in \mathbb{C}$ is the inner product and $|\psi\rangle\langle\psi| = \psi\psi^* \in M_n$ the outer product; in particular, $|i\rangle\langle j| = E_{ij}$. A bipartite *product vector* is a simple tensor $|\psi\rangle \otimes |\varphi\rangle$, also written $|\psi \otimes \varphi\rangle$ or $|\psi\varphi\rangle$, where $\otimes$ denotes the *tensor product*, realized concretely as the Kronecker product of coordinate vectors.

We equip M_n with involution $*$, that is, conjugate transposition ($A^* = \overline{A^T}$) over $\mathbb{C}$ and transposition ($A^* = A^T$) over $\mathbb{R}$. We denote the subspace of Hermitian matrices ($A = A^*$) by $H_n \subseteq M_n(\mathbb{C})$, and the subspaces of symmetric ($A = A^T$) and skew-symmetric ($A = -A^T$) matrices by $S_n, K_n \subseteq M_n(\mathbb{R})$ respectively, so that $M_n(\mathbb{R}) = S_n \oplus K_n$. Finally, $M_n^+ \subseteq H_n$ is the cone of positive semidefinite matrices.

Definition 1.1 *(Positive semidefinite and positive definite)* A matrix $A \in H_n$ is *positive semidefinite* (PSD), written $A \succeq 0$, if $x^* A x \geq 0$ for all $x \in \mathbb{C}^n$, or equivalently if all eigenvalues of A are non-negative. It is *positive definite* (PD), written $A \succ 0$, if $x^* A x > 0$ for all $x \in \mathbb{C}^n \setminus \{0\}$, equivalently if all eigenvalues are strictly positive.
The *Hilbert-Schmidt inner product* on M_n is defined as $\langle A, B \rangle := \operatorname{tr}(B^* A)$, making M_n an inner product space. In the case of H_n or S_n, it simplifies to $\operatorname{tr}(BA)$.

We work with bipartite systems on the tensor product space $\mathbb{C}^m \otimes \mathbb{C}^n$. We equip the operator space $M_m \otimes M_n \simeq M_{mn}$ with multiplication, i.e., for $A, C \in M_m$ and $B, D \in M_n$ we have $(A \otimes B)(C \otimes D) = AC \otimes BD$, where $\otimes$ is the Kronecker tensor product. We use the *partial trace* tr_k, as the generalization of the trace operation, to *trace out* the k-th factor, for example $\operatorname{tr}_1(A \otimes B) = \operatorname{tr}(A)B$.

Definition 1.2 *(Partial transpose)* For a product matrix $A \otimes B \in M_m \otimes M_n$, the *partial transpose* with respect to subsystem B is defined as

$$(A \otimes B)^{\Gamma_B} := A \otimes B^T, \tag{1}$$

and extended by linearity to a general $\rho \in M_m \otimes M_n$.
For Hermitian ρ, the partial transpose with respect to either subsystem yields the same result: $\rho^{\Gamma_A} = \rho^{\Gamma_B}$, so we abbreviate to ρ^{Γ} without loss of generality.

Given linear maps $\Phi : M_n \to M_m$ and $\Psi : M_p \to M_q$, their *tensor product* $\Phi \otimes \Psi : M_n \otimes M_p \to M_m \otimes M_q$ is the linear map defined on simple tensors by

$$(\Phi \otimes \Psi)(A \otimes B) := \Phi(A) \otimes \Psi(B) \tag{2}$$

and extended by linearity. For the identity map I and some linear map Φ, we say $I \otimes \Phi$ is its *ampliation*.

Definition 1.3 *(Positive, k-positive, and completely positive maps)* Let $\Phi : M_n \to M_m$ be a linear map.

- Φ is *positive* (P) if $A \succeq 0 \Rightarrow \Phi(A) \succeq 0$.
- Φ is *k-positive* if $I_k \otimes \Phi : M_k \otimes M_n \to M_k \otimes M_m$ is positive.
- Φ is *completely positive* (CP) if it is k-positive for every $k \in \mathbb{N}$ [12].

Clearly every completely positive map is also positive. However, there are many maps that are positive but not completely positive (PNCP). They will be playing a central role in this work.

Definition 1.4 *(Block positivity)* An operator $W \in H_{mn}$ is *block positive* if

$$\langle x \otimes y | W | x \otimes y \rangle \geq 0 \tag{3}$$

for all product vectors $x \otimes y$ with $x \in \mathbb{C}^m$ and $y \in \mathbb{C}^n$.

Definition 1.5 *(Choi-Jamiolkowski isomorphism)* Let $\Phi : M_n \to M_m$ be a linear map. The ampliation of Φ applied to the maximally entangled state $|\Omega\rangle = \frac{1}{\sqrt{n}} \sum_{i=1}^n |i\rangle \otimes |i\rangle$ yields its *Choi matrix*:

$$C_\Phi = (I \otimes \Phi)(|\Omega\rangle\langle\Omega|) = \frac{1}{n} \sum_{i,j=1}^n |i\rangle\langle j| \otimes \Phi(|i\rangle\langle j|) \in M_n \otimes M_m. \tag{4}$$

The assignment $\Phi \mapsto C_\Phi$ is a linear isomorphism between maps $\Phi : M_n \to M_m$ and matrices in $M_n \otimes M_m$ [25]. The map can be recovered from its Choi matrix by $\Phi(\rho) = n \operatorname{tr}_1\left[(\rho^T \otimes I_m) C_\Phi\right]$.
Equivalently, we can view C_Φ as an $n \times n$ block matrix $C_\Phi = \left[\Phi(E_{ij})\right]_{i,j}$, where $\Phi(E_{ij}) \in M_m$ is the (i,j)-th block.

Theorem 1.6 Under the Choi-Jamiołkowski isomorphism, Φ is completely positive if and only if $C_\Phi \succeq 0$ [26], and Φ is trace-preserving if and only if $\operatorname{tr}_2[C_\Phi] = I_n$.
Block positivity is the operator counterpart of map positivity: a map Φ is positive if and only if its Choi matrix C_Φ is block positive, exactly as it is completely positive if and only if $C_\Phi \succeq 0$.

The partial transpose is itself the ampliation of the transposition map T: $\rho^\Gamma = (I_m \otimes T)(\rho)$. Writing this out for the 2×2 case we get its Choi matrix:

$$C_T = \sum_{i,j=1}^2 E_{ij} \otimes E_{ij}^T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{5}$$

with eigenvalues ± 1. Since $C_T \not\succeq 0$, the transpose is *not* completely positive even though it is positive; it is the prototypical PNCP map.

A word to the wise: the Choi matrix of the composition $\Phi \circ \Psi$ is not the product $C_\Phi C_\Psi$. Instead, it is obtained from

$$C_{\Phi \circ \Psi} = \sum_{i,j} E_{ij} \otimes \Phi\big(\Psi(E_{ij})\big). \tag{6}$$

Reading off C_Ψ in block notation gives $\Psi(E_{ij}) = \sum_{k,l} (C_\Psi)_{ik,jl} E_{kl}$, hence by linearity

$$\Phi\big(\Psi(E_{ij})\big) = \sum_{k,l} (C_\Psi)_{ik,jl} \Phi(E_{kl}). \tag{7}$$

The (p, q) entry of $\Phi(E_{kl})$ is $(C_\Phi)_{kp,lq}$, so the (p, q) entry of block (i, j) of $C_{\Phi\circ\Psi}$ is

$$(C_{\Phi\circ\Psi})_{ip,jq} = \sum_{k,l} (C_\Psi)_{ik,jl} (C_\Phi)_{kp,lq}. \tag{8}$$

This is used directly in the implementation to compose two candidate PPT maps from their Choi matrices.

1.2. Entanglement. A *quantum state* on $\mathbb{C}^n$ is a density matrix $\rho \in M_n^+$ with $\mathrm{tr}(\rho) = 1$. A *quantum channel* is a completely positive trace-preserving (CPTP) map $\Phi : M_n \to M_m$.

Definition 1.7 *(Separability and entanglement)* A bipartite state $\rho \in M_m \otimes M_n$ is *separable* if is is of the form:

$$\rho = \sum_i p_i \rho_i^A \otimes \rho_i^B, \quad p_i \geq 0, \quad \sum_i p_i = 1, \quad \rho_i^A \in M_m^+, \quad \rho_i^B \in M_n^+. \tag{9}$$

Otherwise ρ is *entangled*.

Definition 1.8 *(Entanglement-breaking map)* A map $\Phi : M_n \to M_m$ is *entanglement breaking* (EB) if $(I_k \otimes \Phi)(\rho)$ is separable for every $k \in \mathbb{N}$ and every state $\rho \in M_k \otimes M_n$ [1]. Testing $k = 1$ alone would only check that Φ preserves separability of bipartite states; the ampliation over all k ensures Φ destroys entanglement with any external reference system. Equivalently, Φ is EB if and only if its Choi matrix C_Φ is a separable state [1].

Definition 1.9 *(PPT state and PPT map)* A state $\rho \in M_m \otimes M_n$ is *PPT* if $\rho \succeq 0$ and $\rho^\Gamma \succeq 0$. A map $\Phi : M_n \to M_m$ is *PPT* if its Choi matrix is PPT:

$$C_\Phi \succeq 0 \quad \text{and} \quad C_\Phi^\Gamma \succeq 0. \tag{10}$$

Since $C_\Phi \succeq 0$ characterizes CP maps, a PPT map is a CP map whose Choi matrix has positive semidefinite partial transpose. The composition of two PPT maps is trivially PPT.

1.2.1. PPT2 Conjecture. With PPT maps (Definition 1.9) and entanglement-breaking maps (Definition 1.8) in hand, we can now state the central object of study.

Theorem 1.10 *(PPT2 Conjecture)* If Φ_1 and Φ_2 are PPT maps, then $\Phi_1 \circ \Phi_2$ is entanglement breaking [11].

The conjecture is trivial in 2×2 and 2×3, since the Horodecki criterion applies [5], it has also been proven for 3×3 twice independently [11, 12], and for certain groups of maps even in higher dimensions [13]. However, it remains open for 4×4 and above in the general case. The rest of this section develops the formalism needed to study the conjecture computationally.

1.3. Testing entanglement. Deciding separability is NP-hard in general [2, 3]. Since an exact test is computationally intractable, one relies on necessary conditions: a state that fails such a condition must be entangled. However, satisfying all known criteria does not guarantee separability. For a comprehensive survey see [27].

1.3.1. The PPT test. If a state ρ is separable, i.e., $\rho = \rho_A \otimes \rho_B$, then it is also PPT, since $\rho^\Gamma = \rho_A \otimes \rho_B^T \succeq 0$ [4]. Furthermore, the transposition map T is positive but not completely positive, applying $I \otimes T$ to an entangled state can produce a non-PSD result. This means that if $\rho^\Gamma \not\succeq 0$, then it must be entangled. The condition is necessary but not sufficient: bound entangled (PPT entangled) states exist for systems with $mn > 6$ [6, 28].

1.3.2. Positive maps and entanglement witnesses. The set of separable states SEP is convex and closed. By the Hahn-Banach separation theorem; any point outside a closed convex set is separated from it by a hyperplane, for every $\rho \notin \mathrm{SEP}$ there exists a Hermitian operator W (a hyperplane in H_{mn}) with $\mathrm{tr}[W\rho] < 0$ and $\mathrm{tr}[W\sigma] \geq 0$ for all $\sigma \in \mathrm{SEP}$ [1].

Definition 1.11 *(Entanglement witness)* A Hermitian operator $W \in H_{mn}$ is an *entanglement witness* if $\mathrm{tr}[W\sigma] \geq 0$ for every separable state $\sigma \in M_m \otimes M_n$ and $\mathrm{tr}[W\rho] < 0$ for some entangled state $\rho \in M_m \otimes M_n$.

Geometrically, each witness W cuts off a half-space that contains ρ but not SEP; the intersection of all such half-spaces recovers SEP exactly.

Witnesses are one-sided: $\mathrm{tr}[W\rho] < 0$ certifies entanglement, but $\mathrm{tr}[W\rho] \geq 0$ does not certify separability. An operator is a (valid) entanglement witness precisely when it is block positive but not PSD.

Definition 1.12 *(Decomposable witness)* A witness W is *decomposable* if $W = P + Q^\Gamma$ for some $P, Q \succeq 0$; otherwise it is *non-decomposable* [29].

A known result is that decomposable witnesses cannot detect PPT entangled states. For any PPT σ, we have

$$\mathrm{tr}\big[(P + Q^\Gamma)\sigma\big] = \mathrm{tr}[P\sigma] + \mathrm{tr}[Q\sigma^\Gamma] \geq 0. \tag{11}$$

This means only non-decomposable witnesses are useful for our purposes.

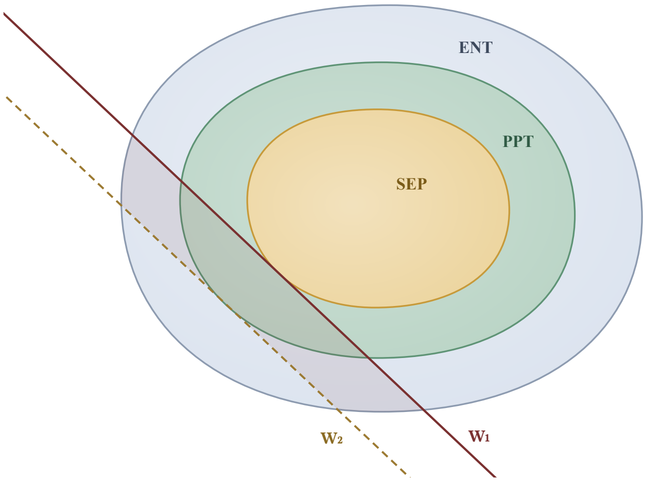

Figure 1: Geometric picture (schematic) of entanglement witnesses as separating hyperplanes over the nested convex sets of separable, PPT and entangled states. Inclusions are strict, with bound-entangled states populating the outside of the PPT region (green strip). The entanglement witness W_1 (solid) is *finer* than W_2 (dashed), it detects every state W_2 does, in addition to the states W_2 misses (shaded band).

Under the Choi-Jamiolkowski isomorphism, every entanglement witness $W = C_\Phi$ corresponds to a PNCP map Φ, and vice versa; block positivity of W is exactly positivity of Φ. This gives two ways to test a state. The scalar test asks whether $\mathrm{tr}[C_\Phi \rho] < 0$, i.e., whether ρ lies on the negative side of the single separating hyperplane defined by C_Φ. The map test asks the more demanding question whether the ampliation $(I_k \otimes \Phi)(\rho)$ has *any* negative eigenvalue, $(I_k \otimes \Phi)(\rho) \not\succeq 0$. The scalar test inspects only one expectation value and so can miss a negative eigenvalue in a direction other than the one C_Φ singles out; the map test inspects the whole spectrum. Concretely, $(I_k \otimes \Phi)(\rho) \not\succeq 0$ holds exactly when some vector ψ satisfies

$$\langle \psi | (I \otimes \Phi)(\rho) | \psi \rangle = \mathrm{tr}[(I \otimes \Phi^*)(|\psi\rangle\langle\psi|)\rho] < 0. \tag{12}$$

The map therefore encodes an entire family of scalar witnesses $W_\psi = (I \otimes \Phi^*)(|\psi\rangle\langle\psi|)$, one per output vector $|\psi\rangle$; the Choi-matrix test $\mathrm{tr}[C_\Phi \rho]$ is the single member obtained from the maximally entangled $|\psi\rangle$, whereas the eigenvalue check optimizes over all $|\psi\rangle$ at once. Hence

$$\lambda_{\min}((I \otimes \Phi^*)(\rho)) \leq \mathrm{tr}[C_\Phi \rho], \tag{13}$$

so the map detects every state the scalar test does, and in general strictly more [1].

The PPT test is the special case of this positive-map test in which $\Phi = T$ is the transposition map: $\rho^\Gamma \not\succeq 0$ is exactly $(I \otimes T)(\rho) \not\succeq 0$. The transpose is, however, a *decomposable* witness, so it sees only *NPT* (non-positive partial transpose, i.e. $\rho^\Gamma \not\succeq 0$) entanglement; detecting PPT entanglement requires stronger maps.

1.3.3. Further separability criteria. Several other necessary conditions for separability are known; each can detect entanglement the PPT test misses, but none is sufficient.

Reduction criterion. If $\rho \in M_m \otimes M_n$ is separable, then

$$\rho_A \otimes I_n - \rho \succeq 0 \quad \text{and} \quad I_m \otimes \rho_B - \rho \succeq 0, \tag{14}$$

where $\rho_A = \mathrm{tr}_B[\rho]$ and $\rho_B = \mathrm{tr}_A[\rho]$ are the *reduced states* obtained by tracing out one subsystem [30].

Range criterion. If $\rho \in M_m \otimes M_n$ is separable, there exist product vectors $\{|\psi_i \otimes \varphi_i\rangle\}$ spanning the range of ρ such that $\{|\psi_i \otimes \overline{\varphi_i}\rangle\}$ spans the range of ρ^Γ [28]. The range criterion can detect certain PPT entangled states, but fails when ρ is full rank (e.g., under noise), since then any set of vectors spans its range trivially.

Majorization criterion. If ρ is separable, then $\lambda(\rho) \preceq \lambda(\rho_A)$ and $\lambda(\rho) \preceq \lambda(\rho_B)$, where $\lambda(\cdot)$ denotes the non-increasingly ordered eigenvalue vector [31]. Note, here $a \preceq b$ has nothing to do with positive definiteness; it denotes *majorization*: $\sum_{i=1}^k a_i \leq \sum_{i=1}^k b_i$ for all k, with equality at $k = n$. This criterion follows from the reduction criterion [31] and therefore shares its limitations.

CCNR / realignment criterion. Define the *realignment* of ρ as the matrix $R(\rho)$ with entries $(R(\rho))_{i(m+k),j(n+l)} = \rho_{i(n+j),k(n+l)}$. If ρ is separable, then $\|R(\rho)\|_1 \leq 1$, where $\|\cdot\|_1$ is the trace norm [32]. The CCNR criterion is independent of the PPT criterion and can detect some PPT entangled states that the other criteria miss.

Beyond criterion-based tests, there are *algorithmic* approaches that reformulate separability as a convex optimization problem, i.e., the DPS hierarchy (Section 1.4.4). There are also criteria based on covariance matrices and Bell inequalities that are less directly applicable in our setting, for a full survey see [27]. The catalogue keeps growing: recent additions include witnesses built from symmetric measurements [33] and mutually unbiased bases [34], realignment-like witness classes [35], and rank-based detection aimed specifically at bound entanglement [36].

1.4. Semidefinite Programming. A *semidefinite program* (SDP) is a convex optimization problem in which a linear objective is minimized subject to a linear matrix inequality. The standard primal form is

$$\begin{aligned} &\text{minimize} && c^T \boldsymbol{x} \\ &\text{subject to} && F(\boldsymbol{x}) := F_0 + \sum_{i=1}^{n} x_i F_i \succeq 0, \end{aligned} \tag{15}$$

where $\boldsymbol{x} \in \mathbb{R}^n$ is the optimization variable, $c \in \mathbb{R}^n$ is the cost vector, and $F_0, F_1, ..., F_n \in H_d$ are fixed Hermitian matrices. The associated *dual SDP* is

$$\begin{aligned} &\text{maximize} && -\operatorname{tr}[F_0 Z] \\ &\text{subject to} && Z \succeq 0, \\ & && \operatorname{tr}[F_i Z] = c_i, \quad i = 1, ..., n. \end{aligned} \tag{16}$$

The *Slater condition* for (15) requires the existence of a strictly feasible point: some $\boldsymbol{x}$ with $F(\boldsymbol{x}) \succ 0$. When it holds for both primal and dual, *strong duality* holds: primal and dual optima coincide. When $c = 0$, (15) is a *feasibility problem*. If infeasible, a dual feasible $Z \succeq 0$ with $\operatorname{tr}[F_i Z] = 0$ and $\operatorname{tr}[F_0 Z] > 0$ certifies infeasibility.

1.4.1. Interior point methods. SDPs are solved in practice by *interior point methods* (IPMs), which follow a smooth trajectory through the strict interior of the feasible region. A standard IPM applied to a $d \times d$ SDP with n variables reaches ε-accuracy in $O\left(\sqrt{d}\log(1/\varepsilon)\right)$ iterations, each requiring $O(n^2 d^2 + n d^3)$ operations [37]. This thesis uses MOSEK [38], a state-of-the-art IPM solver for semidefinite programs.

A property of IPMs important for our use: solutions lie in the *strict interior* of the feasible region $F(\boldsymbol{x}) \succ 0$. Since all solutions are floating-point approximations, this provides the numerical slack around the boundary that enables rationalization as a post solver method of certifying solutions.

1.4.2. Sum-of-squares relaxation. Semidefinite programming also underlies a general relaxation of polynomial non-negativity. Let $\mathbb{R}[\boldsymbol{x}, \boldsymbol{y}]$ be the ring of real polynomials in $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{y} \in \mathbb{R}^m$. A polynomial p is *non-negative* if $p(\boldsymbol{x}, \boldsymbol{y}) \geq 0$ for all real inputs; it is a *sum of squares* (SOS) if $p = \sum_i q_i^2$ for polynomials q_i. Every SOS polynomial is non-negative, but not every non-negative polynomial is SOS. The classic example is the Motzkin polynomial $M(x, y) = x^4 y^2 + x^2 y^4 - 3x^2 y^2 + 1$ [39], non-negative by the AM–GM inequality yet provably not SOS. In general however, such examples are difficult to find. By Hilbert's theorem [40] the two notions coincide only in special cases, and explicit non-negative non-SOS polynomials remained elusive for decades after Hilbert's nonconstructive proof of their existence.

Definition 1.13 *(Gram matrix representation)* Let $\boldsymbol{z} = (z_1, ..., z_N) \in \mathbb{R}^N$ and let $\boldsymbol{v}(\boldsymbol{z})$ represent the $\binom{N+d-1}{d}$ monomials of degree d in $\boldsymbol{z}$. A homogeneous polynomial p of degree $2d$ is SOS if and only if there exists a symmetric PSD matrix $G \succeq 0$ (a *Gram matrix* of p), such that

$$p(\boldsymbol{z}) = \boldsymbol{v}(\boldsymbol{z})^T G \boldsymbol{v}(\boldsymbol{z}). \tag{17}$$

Testing whether p is SOS therefore reduces to a semidefinite feasibility problem: find $G \succeq 0$ satisfying the linear constraints that equate coefficients of (17) with those of p. If no such G exists, p is not SOS.

Testing the nonnegativity of a polynomial p is an NP-hard problem [41]. The standard way around this is to use SOS relaxations which are computationally tractable. The idea is to multiply p by a fixed SOS multiplier and test the product for SOS [42].

1.4.3. Entanglement testing as an SDP. We can rephrase entanglement testing as maximizing $\operatorname{tr}[M\rho]$, where $\rho \in \mathrm{SEP}$. This is a convex optimization problem, but the set SEP is not semidefinite representable. Therefore, we find tractable outer approximations of SEP, to get (a sequence of) necessary conditions for separability. Each approximation is defined by semidefinite constraints, so membership can be tested by an SDP feasibility problem. If ρ fails the test at some level, we can conclude it must be entangled.

1.4.4. The DPS hierarchy. The Doherty-Parrilo-Spedalieri (DPS) hierarchy [19, 20] provides such a sequence of SDP relaxations of separability, with the added benefit of being *complete*: for every entangled state there exists a finite level k at which the test detects it.

In principle, running the hierarchy to convergence solves separability exactly; in practice, only the first few levels are computationally feasible.

Symmetric extensions. For a separable state $\rho = \sum_i p_i \rho_i^A \otimes \rho_i^B$, define the k-fold *symmetric extension*

$$\rho_k = \sum_i p_i \rho_i^A \otimes \left(\rho_i^B\right)^{\otimes k} \in H\left(\mathbb{C}^m \otimes (\mathbb{C}^n)^{\otimes k}\right). \tag{18}$$

This extension satisfies three properties, each with a physical interpretation:

$$(I_m \otimes \Pi_k)\rho_k(I_m \otimes \Pi_k) = \rho_k, \tag{19}$$

$$\rho_k^{\Gamma_s} \succeq 0 \quad \forall s = 1, ..., k, \tag{20}$$

$$\mathrm{tr}_{B_2 \dots B_k}[\rho_k] = \rho. \tag{21}$$

Here Π_k projects onto the bosonic (fully symmetric) subspace of $(\mathbb{C}^n)^{\otimes k}$. Condition (19) states that ρ_k is invariant under permutations of the k copies of B. Condition (20) states that all partial transposes of ρ_k are PSD. Condition (21) states that tracing out the extra $k-1$ copies of B recovers ρ.

Definition 1.14 *(DPS set at level k)* DPS_n^k is the set of states $\rho \in H(\mathbb{C}^m \otimes \mathbb{C}^n)$ for which there exists a symmetric extension $\rho_k \in H\left(\mathbb{C}^m \otimes (\mathbb{C}^n)^{\otimes k}\right)$ satisfying (19), (20), and (21).

Each DPS_n^k is defined by semidefinite constraints, so membership is testable in polynomial time for fixed k. The hierarchy satisfies [20]:

1. $\mathrm{SEP}_n \subseteq \mathrm{DPS}_n^k$ and $\mathrm{DPS}_n^{k+1} \subseteq \mathrm{DPS}_n^k$ for all $k \geq 1$.
2. DPS_n^1 is equivalent to the PPT criterion.
3. Asymptotic completeness: $\bigcap_{k \geq 1} \mathrm{DPS}_n^k = \mathrm{SEP}_n$.

Feasibility and witness extraction. Testing $\rho \in \mathrm{DPS}_n^k$ amounts to searching for ρ_k satisfying (19), (20), and (21): this is an SDP feasibility problem. When it is infeasible (no valid extension exists), the dual variable $Z \succeq 0$ yields an entanglement witness. Concretely, the SDP dual to the level-k test has a feasible Z whenever ρ is entangled at that level. The operator

$$W = \mathrm{tr}_{B_2 \dots B_k}[Z] \tag{22}$$

is an entanglement witness for ρ [20], i.e., $\mathrm{tr}[W\rho] < 0$ verifies entanglement directly.

Improvements and limitations. Several enhancements to the basic hierarchy are known. Harrow, Natarajan, and Wu [43] add first-order optimality (KKT) conditions to the DPS SDP, achieving *finite convergence*: infeasibility is certified at a finite level rather than only asymptotically. The KKT conditions are linear constraints on the Lagrange multipliers of the original SDP; they increase the variable count substantially and make clean witness extraction from the dual more involved. Specialized hierarchies have been developed for states with symmetry. For diagonal unitary invariant states, Britz and Laurent [44] give a drastically smaller SDP at each level. For Werner states and isotropic states, explicit separability conditions are known. However, none of these apply to our search: the PPT2 conjecture is already proven for the relevant symmetric families [13], so our search must use the general DPS hierarchy.

1.4.5. Maps as polynomials. An alternative approach to detecting entanglement exploits the polynomial representation of linear maps. Each linear map $\Phi : M_n \to M_m$ corresponds to a biquadratic polynomial:

$$p_\Phi(\boldsymbol{x}, \boldsymbol{y}) := \boldsymbol{y}^T \Phi(\boldsymbol{x}\boldsymbol{x}^T)\boldsymbol{y}. \tag{23}$$

The fundamental correspondence [21] between the map and polynomial representation is:

- Φ is *positive* if and only if p_Φ is non-negative on $\mathbb{R}^n \times \mathbb{R}^m$.
- Φ is *completely positive* if and only if p_Φ is SOS.

A PNCP map corresponds exactly to a non-negative non-SOS polynomial. Each such a polynomial therefore gives rise to an entanglement witness, and the SOS test of the previous section becomes a test of complete positivity. This is the route we use to manufacture entanglement witnesses directly.

KMŠZ construction for PNCP maps. Now that we have established the value of non-negative non-SOS polynomials for our purposes, we present a construction [21] that produces such polynomials, and therefore PNCP maps, as follows:

1. Sample random points $x^{(1)}, ..., x^{(t)} \in \mathbb{R}^n$ and $y^{(1)}, ..., y^{(t)} \in \mathbb{R}^m$.
2. Form bilinear $h_j(\boldsymbol{x}, \boldsymbol{y}) = \langle x^{(j)}, \boldsymbol{x}\rangle \cdot \langle y^{(j)}, \boldsymbol{y}\rangle$, each a product of two linear forms, so $\sum_j h_j^2$ is SOS.
3. Find $f \notin \mathrm{span}\{h_1, ..., h_t\}$ such that f is not SOS.
4. Find $\delta > 0$ small enough that $F_\delta := \delta f + \sum_j h_j^2 \geq 0$.

The construction of F_δ ensures it is *not* SOS, so all that is left is to verify the non-negativity. Steps 1–3 use only linear algebra, and step 4 involves solving an SDP. However, we cannot directly represent F_δ with semidefinite constraints. Instead, we relax the condition and search for a Gram matrix $G \succeq 0$ for $F_\delta \cdot S$, where $S = \sum_{i,j} x_i^2 y_j^2$ is a fixed SOS multiplier. This bihomogeneous multiplier preserves the separate degrees in $\boldsymbol{x}$ and $\boldsymbol{y}$, matching the biquadratic structure of F_δ. If feasible, then $F_\delta \cdot S$ is SOS, which certifies that F_δ is non-negative [21].

Real maps and their complexification. The correspondence above is stated for real symmetric inputs: $p_\Phi(\boldsymbol{x}, \boldsymbol{y}) = \boldsymbol{y}^T \Phi(\boldsymbol{x}\boldsymbol{x}^T)\boldsymbol{y}$ probes Φ only on rank-one *real* symmetric matrices $\boldsymbol{x}\boldsymbol{x}^T$, and "non-negative" means $p_\Phi \geq 0$ on $\mathbb{R}^n \times \mathbb{R}^m$. Accordingly, the KMŠZ construction produces a *real* linear map

$$\Phi : S_n(\mathbb{R}) \to S_m(\mathbb{R}). \tag{24}$$

Quantum states, however, are complex Hermitian operators, so to use Φ as an entanglement witness it must be extended to a complex map. Following [21], we take the *complexified trivial extension*

$$\Gamma_\mathbb{C} := (\Phi \oplus 0)_\mathbb{C} : M_n(\mathbb{C}) \to M_m(\mathbb{C}), \tag{25}$$

defined in two steps. First, using $M_n(\mathbb{R}) = S_n(\mathbb{R}) \oplus K_n(\mathbb{R})$, extend Φ to all real matrices by acting as zero on the skew-symmetric part:

$$(\Phi \oplus 0)(S + K) = \Phi(S), \quad S \in S_n(\mathbb{R}),\ K \in K_n(\mathbb{R}). \tag{26}$$

Then extend †-linearly to complex matrices,

$$\Gamma_\mathbb{C}(A + iB) = (\Phi \oplus 0)(A) + i(\Phi \oplus 0)(B), \quad A, B \in M_n(\mathbb{R}), \tag{27}$$

the unique extension satisfying $\Gamma_\mathbb{C}(X^*) = \Gamma_\mathbb{C}(X)^*$. Klep et al. prove that $\Gamma_\mathbb{C}$ is positive (respectively PNCP) whenever Φ is [21].

Two observations make this extension harmless to compute with. First, for a Hermitian input $\rho = A + iB$ (with $A \in S_n(\mathbb{R})$ symmetric and $B \in K_n(\mathbb{R})$ skew-symmetric, since $\rho = \rho^*$) only the symmetric part survives, $\Gamma_\mathbb{C}(\rho) = \Phi(A)$. Second, and equivalently, the Choi matrix of $\Gamma_\mathbb{C}$ is exactly the real symmetric matrix C_Φ returned by the construction, now read as a complex Hermitian operator; no entries change. A by-product of annihilating the skew part is that C_Φ is *partial-transpose invariant*, $C_\Phi^\Gamma = C_\Phi$, a property we return to in Section 2.3.3.

Indecomposability of the generated maps. For complex maps the appropriate polynomial object is *bi-Hermitian*: with $z \in \mathbb{C}^n$, $w \in \mathbb{C}^m$ and their conjugates, the polynomial

$$p_{\Gamma_\mathbb{C}}(z, w) = \langle w | \Gamma_\mathbb{C}(|z\rangle\langle z|) | w \rangle \tag{28}$$

is real-valued, and it is non-negative on all product vectors if and only if $C_{\Gamma_\mathbb{C}}$ is block-positive, i.e., an entanglement witness. For such polynomials there are *two* inequivalent notions of sum-of-squares [45]:

- *complex SOS* (CSOS): $p = \sum_i |q_i(z, w)|^2$ with each q_i holomorphic (a polynomial in z, w, not $\overline{z}, \overline{w}$);
- *real SOS* (RSOS): $p = \sum_i g_i^2$ with each g_i a Hermitian (real-valued) polynomial in $z, \overline{z}, w, \overline{w}$.

CSOS is a strictly stronger condition than RSOS. Under the map correspondence [45], a CSOS certificate corresponds exactly to a completely positive map and an RSOS certificate to a *decomposable* witness. Failure of any RSOS decomposition is therefore the polynomial signature of indecomposability, and by Definition 1.12 such a witness can detect PPT entangled states.

The two representations are linked by a structural identity. Writing $z = a + ib$ and $w = c + id$ with $a, b \in \mathbb{R}^n$ and $c, d \in \mathbb{R}^m$, a direct expansion gives

$$p_{\Gamma_\mathbb{C}}(a + ib, c + id) = p_\Phi(a, c) + p_\Phi(b, c) + p_\Phi(a, d) + p_\Phi(b, d). \tag{29}$$

The complexified polynomial is thus completely controlled by the values of the real form p_Φ on real inputs, which can be used to prove *indecomposability* [23].

Theorem 1.15 *(Indecomposability)* If p_Φ is not SOS, then $p_{\Gamma_\mathbb{C}}$ is not RSOS, and hence the witness $C_{\Gamma_\mathbb{C}}$ is non-decomposable.

Since p_Φ is bihomogenous it vanishes whenever either argument is zero, so restricting to the real slice, i.e., setting $b, d = 0$ we find (29) collapses to $p_{\Gamma_\mathbb{C}}(a, c) = p_\Phi(a, c)$ and $p_{\Gamma_\mathbb{C}} \in$ RSOS would imply $p_\Phi \in$ SOS.

Because the KMŠZ form p_Φ is non-SOS by construction, every generated witness is indecomposable, and therefore capable of detecting some PPT entangled state. This settles, at least in principle, our concerns about the efficacy of using such maps for our purposes. Indecomposability is otherwise not easy to come by: general criteria for it are scarce [46], and known indecomposable families are typically hand-crafted [47]. That the construction produces indecomposable witnesses generically and in bulk is one of its main attractions.

Simply knowing the generated witnesses *can* detect PPT entanglement is not our only concern, we also need to know how many such states they can detect. Geometrically, we interpret this as asking where they lie on the cone. We say a witness W is *tight* [48] if there exists some product vector $x \otimes y$ for which $\langle x \otimes y | W | x \otimes y \rangle = 0$, in this case the supporting hyperplane genuinely touches SEP. Further, W is *optimal* [29, 49] if there is no $P \succeq 0$ such that $W - P$ remains block-positive, i.e., *tilting* the hyperplane toward SEP.

By construction, the generated witnesses are tight, when the largest feasible δ is used. The algorithm samples points and prescribes zeros, so the hyperplane defined by W touches SEP at those points. Optimality however, is a stronger condition, and as of yet undetermined for these witnesses [23]. A practical consequence is that the witnesses lie close to the decomposable cone, and while they are robust in detecting NPT entanglement, they detect PPT entanglement only weakly, in the sense that the most negative expectation $\mathrm{tr}[W\rho]$ they attain on a PPT entangled state is tiny in magnitude, so a detection barely crosses zero and is easily lost to noise.

Non-uniqueness of the matrix representation. The polynomial p_Φ does not determine the matrix C_Φ uniquely. Writing p_Φ as a quadratic form in the product monomials $\boldsymbol{z} = \boldsymbol{x} \otimes \boldsymbol{y}$,

$$p_\Phi(\boldsymbol{x}, \boldsymbol{y}) = \boldsymbol{z}^T M \boldsymbol{z}, \tag{30}$$

we see any symmetric M that reproduces the coefficients of p_Φ is an admissible representation. Two such matrices M, M' give the same polynomial precisely when $\boldsymbol{z}^T (M - M')\boldsymbol{z} = 0$. The matrices with this property form a linear space L, spanned by the 2×2 minor (*Segre*) relations

$$(x_i y_k)(x_j y_l) - (x_i y_l)(x_j y_k) = 0, \quad i < j,\ k < l. \tag{31}$$

There is one independent relation for each choice of two rows $i < j$ and two columns $k < l$, so $\dim L = \binom{n}{2}\binom{m}{2}$. The admissible representations therefore form an affine space $M_0 + L$.

The important distinction is that the relations defining L vanish on *real* product vectors, not on complex ones, so members of $M_0 + L$ are not necessarily valid witnesses or even block-positive over $\mathbb{C}$. The canonical choice is the symmetric, partial-transpose-invariant representative (25), where zeroing the skew-symmetric part guarantees positivity. Whether the skew-symmetric directions can be used while preserving positivity over $\mathbb{C}$ remains open.

Rationalization. It is possible to sample over $\mathbb{Q}$ to produce a rational polynomial, but the final step in our construction relies on solving an SDP, which inherently works with floating-point arithmetic. For an exact certificate it is therefore more prudent to rationalize only the final result.

Let G be a numerical solution with $\mu = \min(\mathrm{eig}(G_0)) > 0$ and residual $\varepsilon = \max_i |\langle A_i, G_0\rangle - b_i|$, then for $\mu > \varepsilon$, a rational feasible $\widehat{G}$ can be obtained [50],[51] in two steps.

1. Compute a rational approximation $\widehat{G}$ of the gram matrix G satisfying $\|\widehat{G} - G_0\|^2 + \varepsilon^2 < \mu^2$.
2. Project $\widehat{G}$ back to the affine subspace given by the constraints.

With a rational gram matrix $\widehat{G}$ in hand, the coefficients of $\widehat{F}_\delta$ can be isolated using exact computations.

2. Methods

This section defines the computational workflow used in the current implementation. The goal is reproducibility with mathematical traceability. Each computational step corresponds to a defined object or operation from the preceding section. All of the code, command-line drivers, generated datasets, and experiments described in this thesis are openly available in the accompanying repository at https://github.com/noahnovsak/masters-thesis (a Julia package released under GPL-3.0-or-later); the implementation notes below point into it directly, and the Appendix follows a single generated witness from construction through both tests of the conjecture as a concrete illustration.

A note on normalization is in order. A quantum state is conventionally a PSD operator with unit trace, and a quantum channel is trace preserving. The tests in this thesis are, however, positivity and sign based. Detecting entanglement amounts to checking the sign of a partial transpose, a witness expectation, or an SOS margin, none of which depend on the magnitude of the trace. We therefore work throughout with unnormalized operators, and normalize only when necessary. This avoids redundant rescaling in the inner loops without affecting any conclusion.

Two design constraints shape the workflow:

1. *exact* separability testing is intractable in the dimensions where the conjecture is open, partial testing is very much doable, but every complete test is a one-sided relaxation;
2. SDP solutions are floating-point and fragile near feasibility boundaries, so positive detections are re-validated exactly.

2.1. Generating a witness library.

Building a library of PNCP witnesses is the first thing the pipeline does, and a contribution in its own right: being able to generate witnesses quickly and in bulk is what prompted this line of research, and the library is reused throughout. We run the KMŠZ construction to produce a number of PNCP maps, and rationalize each certificate so that every stored witness is exact, guarding against the false positives that floating-point SOS tests invite near the boundary.

The rationalization, developed in Section 1.4.5.5, proceeds as follows:

1. Extract the Gram matrix G from the SDP constraints,
$$(\delta f + h^2)\left(\sum_{i,j} x_i^2 y_j^2\right)^l = (\boldsymbol{xy})^T G \boldsymbol{xy} \tag{32}$$
.
2. Eigendecompose G exploiting the known e-dimensional nullspace [21]; set the first $e = (n-1)(m-1)$ eigenvalues to zero, and rationalize the remaining coefficients to obtain $\hat{G}$.
3. Reformulating the constraints in terms of $\hat{G}$, isolate the coefficients of the polynomial $\hat{p} = \delta f + h^2$ from the relaxation terms $\left(\sum_{i,j} x_i^2 y_j^2\right)^l$.
4. Additionally we solve a final feasibility SDP to double check we have not accidentally made $\hat{p}$ a SOS.

Upon completion we are confident in the validity of the certificate, so we can choose to represent it in floating point or as a rational number, as long as we are sure to use high enough precision. Typically, solvers work with a tolerance of $\approx 10^{-8}$.

2.2. Generating candidates.

To search for a PPT2 counterexample we compose two PPT maps and test the composite channel for entanglement; a candidate is therefore a PPT channel, taken as the Choi matrix of a PPT map. We generate candidates three ways: by generic random sampling, by symmetry-restricted random sampling, and by witness-guided construction. The random sampler produces a PPT state that an entanglement filter keeps only if it can plausibly yield a counterexample; the witness-guided generator instead targets bound entangled states directly, one per witness.

2.2.1. Random PPT states.

We generate random PPT states in two steps:

1. Sample a real matrix $R \in M_{nm}$ with i.i.d. standard-normal entries and form the PSD matrix $\rho = RR^T$.
2. Compute the smallest eigenvalue of the partial transpose, $\lambda = \min(\mathrm{eig}(\rho^\Gamma))$, then set $\rho \leftarrow \rho - \lambda I$, for $\lambda < 0$

The construction is correct without any further work. By definition $\rho = RR^T \succeq 0$, and when $\lambda < 0$ the shift adds the non-negative multiple $|\lambda|\ I$ to ρ, so positivity is preserved. The same shift acts on the partial transpose as $\rho^\Gamma - \lambda I$, because $I^\Gamma = I$, which raises every eigenvalue of ρ^Γ by $|\lambda|$ and hence makes $\rho^\Gamma \succeq 0$. The resulting ρ is therefore PPT. Sampling element-wise from the normal distribution produces a representative collection of states [52].

The off-diagonal blocks may optionally be symmetrized before the shift, for each block pair (i, j) with $i < j$, both ρ_{ij} and ρ_{ji} are replaced by their average $\left(\rho_{ij} + \rho_{ji}\right)/2$, which forces $\rho = \rho^\Gamma$, invariant under partial transpose (IPT). Restricting to such IPT representatives is a permissible convenience rather than a requirement. We only check if it makes the computation any simpler or faster, since a single positivity shift then certifies both $\rho \succeq 0$ and $\rho^\Gamma \succeq 0$ at once. The motivation is that our precomputed witnesses are themselves generated in this IPT shape, so matching the candidates to the same form may make them easier to detect; we therefore run the search in two versions, with and without symmetry.

A random PPT state may be separable or bound entangled, and only the entangled ones are useful here. If either composed map is entanglement breaking, equivalently, has a separable Choi matrix, then the composition is automatically entanglement breaking and cannot violate the conjecture. We therefore discard separable candidates and keep only states for which we can certify entanglement, assembling a pool of genuine bound entangled states whose compositions are worth testing.

2.2.2. Bound entangled states from a witness.

The random sampler has no control over entanglement, so many of its draws are discarded. The witness-guided generator instead targets entanglement directly, using the witness library to manufacture states it is guaranteed to detect. Fix a witness W and minimize its expectation over the whole PPT cone,
$$\begin{aligned} \text{minimize} \quad & \mathrm{tr}[W\rho] \\ \text{subject to} \quad & \rho \succeq 0, \\ & \rho^\Gamma \succeq 0, \\ & \mathrm{tr}[\rho] = 1. \end{aligned} \tag{33}$$

This is a single SDP. Since every separable state gives $\mathrm{tr}[W\rho] \geq 0$, a negative optimum exhibits a PPT *entangled* state that W detects, one bound entangled candidate per witness that admits one. The optimum also measures the witness's detection strength, and is a ready-made hard instance for the following composition tests.

2.2.3. Alternative bound-entangled constructions. The first method, i.e., random sampling is fast, but it gives no control over entanglement. So we must filter states out, which necessitates solving an SDP. Not a great option for scalable searches. The second method is more targeted, but still suffers from the same problem, though the SDP is much smaller in this case. There are several constructions in the literature that produce bound entangled states by design, however we opted not to use them in the search, for various reasons noted below. They are however, the natural starting points for further investigation.

- Unextendible product bases (UPB) are a set of mutually orthogonal product vectors spanning a proper subspace whose orthogonal complement contains no product vector. The normalized projector onto that complement is PPT and entangled [53]. UPBs give explicit, low-rank bound entangled states, but the construction is the opposite of generic: it relies on specific, hand-picked bases that cannot be sampled at random, and in some dimensions may not exist at all, so it cannot drive a randomized search.
- Antisymmetric-subspace states are a simple class of PPT entangled states from the projectors onto the symmetric and antisymmetric subspaces of two identical systems, generalizing the Werner states [54]. The construction is explicit, but certifying entanglement of the resulting states still reduces to an SDP, so it does not scale better than our pipeline.
- Symmetric random induced states are a more recently studied class of bound entangled states [55]. Sampling a random pure state from the symmetric subspace $\mathcal{H}_S^{N+N_A+1}$ – the permutation-invariant, or *Dicke*, subspace of $N + N_A + 1$ qubits, spanned by the Dicke states [56] – and tracing out N_A subsystems, leaving a mixed state on $\mathcal{H}_S^{N+1}$ produces, with high probability, a bound entangled state. The catch is dimensionality: for $N = 4$ bound entanglement is most likely at $N_A = 12$, which demands an enormous ancillary space and correspondingly heavy computation. Even then the probability of entanglement stays below 0.5, comparable to the hit rate of our own construction. It is nonetheless a promising state-of-the-art source of candidates and a useful point of comparison.

2.3. Testing candidates. We keep generation and testing separate so that each pool is built once, checkpointed, and can be reused.

2.3.1. Screening a composite for entanglement. Given two PPT candidates Φ_1, Φ_2 from the pool, we form the Choi matrix of their composition using the ampliation operation (6). Then we test the composite for entanglement: if any composition is ever found to be entangled, the PPT2 conjecture is violated. Because composition is not commutative, the search ranges over *every ordered pair* of pool states, self-pairs included.

We use three distinct criteria to screen the composite channels for entanglement, recording all three so the detectors can be compared directly:

- **Scalar witness test.** For each witness W in the library, $\mathrm{tr}[WC] < 0$ certifies that C is entangled.
- **Map witness test.** The *stronger* condition $(I \otimes \Phi_W)(C) \not\geq 0$, evaluated as a smallest-eigenvalue check on the ampliation of the witness map; this can fire where the scalar test does not.
- **DPS relaxation.** The level-2 DPS relaxation; a feasible dual certificate flags entanglement.

Algorithm 1: PPT2 composition search

Build a pool of candidate PPT maps $\Phi_1, ..., \Phi_N$
for every ordered pair (Φ, Ψ) **do**
  Form the composite channel $C_{\Phi\circ\Psi}$
  for every witness W in the library **do**
    if $\mathrm{tr}[WC_{\Phi\circ\Psi}] < 0$ **then**
      Flag $C_{\Phi\circ\Psi}$ as entangled by the scalar witness test.
    end
    if $(I \otimes \Phi_W)(C_{\Phi\circ\Psi}) \not\geq 0$ **then**
      Flag $C_{\Phi\circ\Psi}$ as entangled by the map witness test.
    end
  end
  if $C_{\Phi_1\circ\Phi_2} \in \mathrm{DPS}^2$ **then**
    Flag $C_{\Phi\circ\Psi}$ as entangled by the DPS relaxation.
  end

15: **end**

2.3.2. Direct PPT2 search by see-saw. The pipeline above tests a finite pool of compositions. We can instead search the composition manifold directly, asking whether a witness W can be made to fire anywhere on it. Sharpening (33), we minimise W's expectation not over the whole PPT cone but over channels that are themselves compositions of two PPT maps:

$$\begin{aligned} \text{minimize} \quad & \operatorname{tr}\left[W C_{\rho_1 \circ \rho_2}\right] \\ \text{subject to} \quad & \rho_{1,2} \succeq 0, \\ & \rho_{1,2}^{\Gamma} \succeq 0, \\ & \operatorname{tr}\left[\rho_{1,2}\right] = 1. \end{aligned} \tag{34}$$

A negative optimum here would exhibit a PPT entangled composition, i.e., a PPT2 counterexample witnessed by W. However, the ampliation is bilinear in the pair (ρ_1, ρ_2), so (34) is no longer a convex optimization problem. We cannot formulate it as a single SDP, instead we have a bilinear matrix inequality, which we solve by *see-saw*, in the spirit of earlier global-optimization searches for witnesses [57]: freezing one factor turns each half-step into an SDP, and we alternate between the two, restarting from several random initialisations. Being non-convex, there is no way to guarantee we reach a global optimum, unlike the certified optimum an SDP returns, but what we do know is that the PPT-cone problem (33) is its convex relaxation and lower bound. This alternating-SDP scheme is only one option among many: we could instead abandon the SDP route entirely and minimise the bilinear objective directly with stochastic gradient descent or one of its extensions, the workhorse of non-convex optimization in machine learning, which might search the manifold more effectively. We leave such alternatives to future work.

2.3.3. A note on the representation freedom. As a by-product we explored the representation freedom of Section 1.4.5.4. Each witness can be expanded into a family of Gram representatives $M_0 + \sum_\alpha \lambda_\alpha N_\alpha$ over a basis $\{N_\alpha\}$ of L. We do not, in the end, rely on this. The relations spanning L vanish on *real* product vectors only, so an asymmetric representative stays block positive over $\mathbb{R}$ but need not remain so over $\mathbb{C}$. Guaranteeing such behavior is to our knowledge also an NP-hard problem without a simple solution in sight. The canonical, IPT representative, annihilating the skew-symmetric part as previously proposed remains the preferred approach for our search, together with generic, non-symmetrized candidates. We leave a principled complex-domain extension to future work.

2.4. Implementation Architecture. The implementation is a small Julia package in the `code/` directory of the repository linked above. The core logic lives in the `ppt2` module (`code/src/ppt2.jl`), with the PNCP construction split into `code/src/pncp.jl` and included into the same module. Together they map one-to-one onto the operations defined above:

Function	Role
`rand_ppt`	Sample a random PPT state per Section 2.2.1, with optional block symmetrization (`ppt_invariant` parameter).
`ampliation`	Compute $(I \otimes A)(B)$, used for map composition and to apply a witness test.
`sample_pncp_form`, `segre_kernel_basis`, `non_sos_form`	The KMŠZ construction: sample random points, build the linear forms h_j, and produce the non-SOS quadratic form f.
`solve_sos`	Set up and solve the SOS feasibility/optimization SDP for a given relaxation degree l; optionally trigger rationalization.
`rationalize_certificate`	Post-solver rationalization: zero the first e Gram eigenvalues, recover rational coefficients, and re-check non-SOS.
`find_pncp_poly`, `pncp_mat`, `poly2mat`	Orchestrate witness generation with retries and export the certificate as a Choi matrix.
`min_ppt_witness`, `min_ppt2_witness`	The witness-restricted SDPs (33) and (34): bound entangled states from a witness, and the see-saw search over compositions.

The module also exports a few supporting primitives: `rand_sep` and `rand_psd` for reference states, `is_ppt` for the PPT check, `gram_freedom` and `is_block_positive` for the witness-representation freedom and the block-positivity check of Section 1.4.5.4, and `swap`/`antisymmetric_projector` for the antisymmetric-subspace construction of [54].

The package leans on the established Julia optimization and quantum-information stack rather than reimplementing it. Polynomials and the SOS cone are handled by `DynamicPolynomials` and `SumOfSquares`; the resulting semidefinite programs are modelled with `JuMP` and solved by MOSEK through `MosekTools` (any `JuMP`-compatible SDP solver could be substituted). The DPS hierarchy is not reimplemented: the search driver calls `entanglement_robustness` from `Ket`, an existing quantum-information toolbox that also supplies utilities such as the partial transpose. Matrices and witness libraries are serialized with `JLD2` so that generation and search can be separated and resumed.

Several command-line drivers in `code/scripts/` orchestrate the long-running jobs, all sharing a `common.jl` harness that provides resumable, reproducible, multithreaded batch generation: completed batches are detected and skipped on a rerun, and every candidate is seeded deterministically so a configuration yields the same dataset regardless of thread count. The generation drivers build the library and the candidate pools: `gen_pncp.jl` constructs the PNCP witness library; `gen_witness_ppt.jl` produces, for each witness, the bound entangled state the witness-restricted SDP extracts from the full PPT cone; and `compare_detection.jl` samples a random bound entangled pool – generic or, with `--ppt-invariant`, symmetry-restricted – while recording every criterion's score on each state, so the detectors can be compared directly (a lighter `gen_ppt.jl` produces such a pool without the scores). `test_ppt2.jl` runs the threaded all-pairs composition search of Algorithm 1, logging any detection together with the offending state and witness; `gen_witness_ppt2.jl` runs the see-saw that sharpens the witness-restricted SDP from the whole PPT cone down to the composition manifold; and `cross_trace.jl` and `cross_ampl.jl` measure how broadly each witness reaches beyond its own state. The `code/test/` suite checks the construction against reference values and verifies that generated maps are positive on large random samples, and the `code/notebooks/` directory documents the rationalization, PPT-state, and UPB workflows interactively.

3. Results

This section reports the final 4×4 scan, the smallest dimension in which the PPT2 conjecture is open. It runs along three threads. First, the PNCP witness library: generating these provably indecomposable witnesses quickly and in bulk is a result in its own right, and the engine behind everything that follows. Second, candidate generation: to look for a counterexample for the PPT2 conjecture we compare three ways of producing PPT states capable of producing a composite channel. Third, the conjecture itself, attacked in two independent ways: by testing the compositions of our candidates, and by a see-saw SDP that moves from a completely random search to something more *optimized.* Every stage was run once under a single fixed seed, at DPS level 2 and tolerance 10^{-8}, so the dataset is reproducible. The full datasets and run logs are archived in the repository under `code/results/`, and the Appendix follows one generated witness from its polynomial through both tests, with the concrete matrices. The headline is a uniform negative: no composition of two PPT maps was ever found entangled, leaving the conjecture without a counterexample.

3.1. The PNCP witness library. The pipeline opens by building a library of PNCP witnesses with the KMŠZ construction, and being able to do so cheaply is the observation that set this work in motion. We generated 10,000 witnesses *in less than an hour*, every construction yielding a valid, rationalized certificate.

Two choices account for the speed. Where the earlier MATLAB prototype [22] drew random *integer* matrices – cosmetically clean, but not generic enough, so it often failed and had to recompute – we sample from a normal distribution and almost always succeed on the first attempt. And where the prototype rationalizes the whole problem up front, paying for rational arithmetic throughout, we rationalize only after the SDP is solved. The most directly comparable recent result [23] builds 20,000 witnesses by the same construction in the *easier* 3×3 case and reports it taking "several days". Ours is faster by orders of magnitude despite the larger dimension (even accounting for our higher computational resources). With these findings, witness generation may no longer be the bottleneck it once was.

3.2. Generating PPT candidates. A PPT2 counterexample is a pair of PPT maps whose composition is entangled. Every candidate is a PPT channel, taken as the Choi matrix of a PPT map. We produce candidates three ways: two random samplers, and one extraction from the witness library. Their comparison in Table 1 is itself one of our findings.

Random sampling by Section 2.2.1 draws a PPT channel and keeps it only if we can certify it entangled. The yield depends sharply on the sampling shape. Of the generic, non-symmetrized draws we could only certify about half as many compared to the symmetrized, partial-transpose-invariant draws. Keep in mind, these are *detection* rates, not true entanglement frequencies: a draw we fail to certify may still be bound entangled but beyond the reach of DPS level 2, so each figure is a lower bound on how often the sampler lands on an entangled state. Even as lower bounds the gap is large, and the symmetric rate exceeds that of the recently studied symmetric random-induced construction of [55], whose entanglement probability stays below one half even at its most favourable ancilla dimension, which is itself large enough to be a real computational bottleneck.

The third way spends no samples. The witness-restricted SDP (33) extracts from each witness a single bound entangled state that witness is guaranteed to detect. The trade is generality for cost: every such state is guaranteed entangled and essentially free to make, but each is tailored to a single witness, as the next section makes plain.

Method	Entangled	Time
Random PPT	32.2% (5,000 / 15,534)	$\approx$ 7 h [1]
Random IPT	59.5% (5,000 / 8,400)	4.5 h
Witness SDP	100% (10,000)	3.6 m

Table 1: The three candidate generators. *Entangled* represents the fraction of produced states we could certify entangled by $\mathrm{DPS}^2_{4\times4}$. Since the hierarchy may miss entangled states this is only a lower bound on the true prevalence. We generated one state per witness in the library and ran the random samplers until 5,000 valid states were found. The main takeaway is the efficiency with which a bound entangled state can be extracted. [1] Generated across several interrupted sessions, so the logged time is unreliable; estimated about 50% above the symmetric run.

3.3. Detecting entanglement. Three criteria are in play: the two witness criteria, the cheap scalar (inner product) test $\mathrm{tr}[W\rho] < 0$ and the stronger map (ampliation) test $(I \otimes \Phi_W)(\rho) \not\succcurlyeq 0$, as well as the level-2 DPS relaxation. Generating candidates and recording every criterion's score lets us compare them directly.

While we can generate witnesses in bulk, we find them essentially single-state detectors. Evaluating our library on the two random pools yielded nothing. Furthermore, cross evaluating them on the witness-derived states, that is 10^8 trials, showed that each witness detects only the state extracted from it.

This leaves us with the DPS hierarchy as the more general detector. An unsurprising result, since the hierarchy has been the de facto standard for separability testing for two decades [20]. But that generality is expensive compared to the witness criteria. The scalar test is a few inner products per state. A negligible cost, growing so gently in the number of witnesses that the library could be enlarged by orders of magnitude at little expense. The map test on the other hand requires an eigen-decomposition of a 16×16 ampliation. At 10,000 witnesses the per-state cost begins to rival a single level-2 DPS solve, which as established is strictly more powerful. So the *stronger* witness criterion costs about as much as the method that dominates it, meaning the only criterion cheap enough to scale may be the weaker one.

The conclusion is blunt: a precomputed witness library is not worth building for general entanglement detection. It reaches nothing on generic states that DPS does not, and the variant that could scale is too weak to matter. Its value is narrow and specific, detecting only the very states it constructs. For the conjecture test we therefore resort to DPS on random states, keeping the witnesses for more targeted approaches.

3.4. Testing the PPT2 conjecture. We test the conjecture two independent ways, neither of which finds a counterexample (see Table 2).

3.4.1. Composing the candidates. The direct route forms the composite of each candidate pair and tests it for entanglement. It is throttled by the quadratic blow-up: a pool of 5,000 states has $5000^2 = 2.5 \times 10^7$ ordered pairs, and at a DPS solve apiece an exhaustive sweep would run for over a year. We therefore test only smaller batches. For example, a 100-state slice of each pool yields $100 \times 100 = 10000$ ordered pairs. Computing across all three families yields 30,000 composite channels - DPS flagged none. Every composition of two PPT maps passed the test well within tolerance. This sweep is necessarily partial. It certifies only the slice it reaches, which is exactly why the second route carries the argument.

3.4.2. Searching the manifold directly.

Rather than test a fixed pool, the see-saw SDP asks whether a witness can be made to fire *anywhere* on the composition manifold. For a fixed W it minimises $\mathrm{tr}\left[WC_{\rho_1\circ\rho_2}\right]$ over composites of two PPT maps; a negative optimum would be a counterexample witnessed by W. The objective is bilinear, hence non-convex. Despite that limitation, two things make it the more compelling route. Its constraints yield a much smaller SDP than the DPS hierarchy leading to substantially faster solves, and it replaces the essentially *spray and pray* approach of the composition scan with a targeted optimization problem. Additionally, an interesting point is that it comes with a clean lower bound: the PPT-cone problem (33), is exactly its convex relaxation.

Over the full PPT cone every one of the 10,000 witnesses is *live*, attaining a negative optimum $\mathrm{tr}[W\rho]$ from -3.5×10^{-2} to -6.1×10^{-8} (median -5.3×10^{-5}). The small typical margins sit right against the decomposable cone, matching the $\approx 10^{-6}$ PPT-violation margins reported for the same construction by [23]. Yet once the state is constrained to the composition manifold, not one witness detects anything. Every see-saw optimum was non-negative, from 1.557×10^{-16} to 2.237×10^{-9} (median 2.985×10^{-10}), flush against the separable boundary but never once crossing it, not even from numerical noise.

This contrast is the central result. Witnesses demonstrably *live* over the full PPT cone go uniformly *dead* once the state must be a composition of two PPT maps, exactly the signature expected if PPT2 holds in 4×4. This is hardly conclusive evidence, but we choose to interpret it as a strong signal that the conjecture is not as surely false as previously believed, and that proving so will take a bit more work.

Method	**Minimum**	**Time**
Random PPT + DPS	6.2×10^{-3}	4.5 h
Random IPT + DPS	4.2×10^{-3}	4 h
Witness SDP + DPS	1.1×10^{-7}	3.5 h
See-saw SDP	1.6×10^{-16}	1.5 h

Table 2: Comparison of independent routes to a PPT2 verdict. The first three test compositions of candidates from a slice of the generated pool against $\mathrm{DPS}^2_{4\times4}$. The last solves a see-saw SDP directly for each witness in the library. All scores are strictly positive, robust even to numerical noise, so no counterexample was found.

3.5. Performance Notes.

A brief note on the computational pitfalls of Hermitian matrices: it may seem reasonable to look for a counterexample in the complex domain, on the intuition that admitting imaginary entries can only enlarge the search and thus help. The issue is that the dimension of the search space doubles, and the SDP size grows accordingly. Our brief experiments showed a slowdown of $\times\, 38$ when computing the DPS relaxation. The much smaller SDPs of the witness-restricted search are more forgiving, and allow for a complex extension. However, we find experimentally that the results are all essentially real. Where this property comes from is unclear at this time, but we use it to project to the real domain and avoid the complex slowdown. It may be an interesting question for future work whether the complex domain can be exploited in some way, and if it is a necessary property for finding a counterexample; the relationship between positivity and entanglement over real and complex Hilbert spaces is itself a subtle and actively studied question [58].

Throughout, our computations are DPS-bound. Every stage that runs the level-2 relaxation: the two random-pool filters, and the witness-pool comparison, is dominated by it, while the witness scans on the same states are comparatively free. Taken together the timings point one way. Random search does not scale. The candidate space in 4×4 is far too large to cover by sampling, the per-pair composition cost rules out an exhaustive sweep, and the witnesses do not generalise to rescue a sparse search. What was fast and decisive was everything *targeted*: states built from witnesses, and the see-saw optimising directly on the manifold. Further searches are therefore better off structured than random: the space is too large to check at random, but with some structure the optimization route looks genuinely promising.

3.6. Complexity and Practical Limits.

All the problems we are looking at are SDPs with exponential [2] complexity in dimension and relaxation depth [20]. Practical bottlenecks include:

1. Processing power: these problems are by nature not easily parallelizable, the interior-point methods used to solve SDPs are intrinsically sequential and rely on factoring large *sparse* matrices, so the potential of GPU speedups is fairly low [38].
2. Memory growth: the size of the SDP grows exponentially with dimension and relaxation depth, leading to memory bottlenecks even for moderate dimensions, i.e., for 4×4 states DPS level 3 is already infeasible on standard hardware, requiring hundreds of gigabytes

of RAM. Even improvements such as adding KKT constraints to achieve finite convergence [43] lead to significant increases in problem size, so they are not without cost.
3. Solver instability near degeneracy: SDP solvers produce floating-point solutions, and when the problem is near the boundary of feasibility, small numerical errors can lead to incorrect conclusions about positivity. This is particularly problematic for our purposes, since we are interested in the very existence of such bound states.
4. Although the generated witnesses are provably non-decomposable, they still lie close to the decomposable cone, and as such detect PPT entanglement only weakly. This compounds the numerical fragility of point 3, and how best to improve a given witness, remains open.
5. Searching for PPT candidates in the first place is non-trivial, and random generation may not be sufficient to find counterexamples if they exist. We may require a more structured approach if the volume of the search space is in fact 0 (this would not mean the conjecture holds).

3.7. Future work. Several directions remain open. The most immediate, for building a witness library that detects many entangled states, is the *complex-domain extension of the representation freedom* of Section 1.4.5.4. The asymmetric Gram representatives are valid witnesses over $\mathbb{R}$ but lose block-positivity over $\mathbb{C}$, since the Segre relations spanning L vanish only on real product vectors. Beyond simply annihilating the skew-symmetric part [21], an analytic guarantee for the asymmetric family, or a projection restoring complex block-positivity while keeping its sharper cut of the entangled region, would turn the freedom into a usable supply of independent witnesses. We pursued this only numerically and found the evidence too weak to rely on.

A second direction toward a counterexample is *structured candidate generation*. Random sampling of the candidate maps and witnesses cannot cover the search space to any computable extent, and if the volume of bound entangled compositions is in fact vanishing, as the generic separability of composed random PPT states suggests [18], such a search will never reach a counterexample even if one exists. It may instead require constructions beyond ours or those surveyed in Section 2.2.3, such as the extremal PPT entangled states of [59, 60].

Bibliography

1. R. Horodecki, P. Horodecki, M. Horodecki, K. Horodecki, Quantum entanglement. *Rev. Mod. Phys.* **81** (2009).
2. S. Gharibian, Strong NP-hardness of the quantum separability problem. *Quantum Inf. Comput.* **10** (2010).
3. L. M. Ioannou, Computational complexity of the quantum separability problem. *Quantum Inf. Comput.* **7** (2007).
4. A. Peres, Separability criterion for density matrices. *Phys. Rev. Lett.* **77** (1996).
5. M. Horodecki, P. Horodecki, R. Horodecki, Separability of mixed states: necessary and sufficient conditions. *Phys. Lett. A* **223** (1996).
6. M. Horodecki, P. Horodecki, R. Horodecki, Mixed-state entanglement and distillation: is there a “bound” entanglement in nature?. *Phys. Rev. Lett.* **80** (1998).
7. K. Horodecki, M. Horodecki, P. Horodecki, J. Oppenheim, Secure key from bound entanglement. *Phys. Rev. Lett.* **94** (2005).
8. M. Christandl, R. Ferrara, Private states, quantum data hiding, and the swapping of perfect secrecy. *Phys. Rev. Lett.* **119** (2017).
9. K. F. Pál, G. Tóth, E. Bene, T. Vértesi, Bound entangled singlet-like states for quantum metrology. *Phys. Rev. Res.* **3** (2021).
10. M. B. Ruskai, M. Junge, D. Kribs, P. Hayden, A. Winter, “Operator structures in quantum information theory” (2012).
11. M. Christandl, A. Müller-Hermes, M. M. Wolf, When do composed maps become entanglement breaking?. *Ann. Henri Poincaré* **20** (2019).
12. L. Chen, Y. Yang, W.-S. Tang, Positive-partial-transpose square conjecture for n = 3. *Phys. Rev. A* **99** (2019).
13. S. Singh, I. Nechita, The PPT² conjecture holds for all Choi-type maps. *Ann. Henri Poincaré* **23** (2022).
14. S. Singh, I. Nechita, Diagonal unitary and orthogonal symmetries in quantum theory II: evolution operators. *Quantum* **5** (2021).
15. S. Prudhoe, Cartan-covariant quantum channels and the PPT² conjecture. *arXiv:2501.03959 [quant-ph]* (2025).
16. M. Kennedy, N. A. Manor, V. I. Paulsen, Composition of PPT maps. *Quantum Inf. Comput.* **18** (2018).
17. M. Rahaman, S. Jaques, V. I. Paulsen, Eventually entanglement breaking maps. *J. Math. Phys.* **59** (2018).
18. B. Collins, Z. Yin, P. Zhong, The PPT square conjecture holds generically for some classes of independent states. *J. Phys. A: Math. Theor.* **51** (2018).
19. A. C. Doherty, P. A. Parrilo, F. M. Spedalieri, Distinguishing separable and entangled states. *Phys. Rev. Lett.* **88** (2002).
20. A. C. Doherty, P. A. Parrilo, F. M. Spedalieri, Complete family of separability criteria. *Phys. Rev. A* **69** (2004).
21. I. Klep, S. McCullough, K. Šivic, A. Zalar, There are many more positive maps than completely positive maps. *Int. Math. Res. Not.* (2019).
22. A. Bhardwaj, A practical approach to SOS relaxations for detecting quantum entanglement. *J. Optim. Theory Appl.* **198** (2023).
23. G. Massé, *et al.*, Detecting bipartite entanglement with PnCP maps and non-negative polynomials. *arXiv:2605.30426 [quant-ph]* (2026).
24. R. Jin, Investigation of the PPT squared conjecture for high dimensions. *arXiv:2010.15554 [quant-ph]* (2020).
25. G. Homa, A. Ortega, M. Koniorczyk, Choi representation of completely positive maps in brief. *Z. Naturforsch. A* **79** (2024).
26. M.-D. Choi, Completely positive linear maps on complex matrices. *Linear Algebra Appl.* **10** (1975).
27. O. Gühne, G. Tóth, Entanglement detection. *Phys. Rep.* **474** (2009).
28. P. Horodecki, Separability criterion and inseparable mixed states with positive partial transposition. *Phys. Lett. A* **232** (1997).
29. M. Lewenstein, B. Kraus, J. I. Cirac, P. Horodecki, Optimization of entanglement witnesses. *Phys. Rev. A* **62** (2000).
30. M. Horodecki, P. Horodecki, Reduction criterion of separability and limits for a class of distillation protocols. *Phys. Rev. A* **59** (1999).
31. M. A. Nielsen, J. Kempe, Separable states are more disordered globally than locally. *Phys. Rev. Lett.* **86** (2001).
32. K. Chen, L.-A. Wu, A matrix realignment method for recognizing entanglement. *Quantum Inf. Comput.* **3** (2003).
33. K. Siudzińska, Indecomposability of entanglement witnesses constructed from symmetric measurements. *Sci. Rep.* **12** (2022).
34. J. Bae, A. Bera, D. Chruściński, B. C. Hiesmayr, D. McNulty, How many mutually unbiased bases are needed to detect bound entangled states?. *J. Phys. A: Math. Theor.* **56** (2023).
35. V. Jannesary, V. Karimipour, D. Chruściński, A class of entanglement witnesses and a realignment-like criterion. *arXiv:2308.07019 [quant-ph]* (2023).
36. A. Gulati, Rank-based entanglement detection for bound entangled states. *Phys. Rev. A* **113** (2026).
37. L. Vandenberghe, S. Boyd, Semidefinite programming. *SIAM Rev.* **38** (1996).
38. MOSEK ApS, The MOSEK optimization toolbox. (2024). Available at: https://www.mosek.com/.
39. T. S. Motzkin, “The arithmetic-geometric inequality” in *Inequalities (Proc. Sympos. Wright-Patterson Air Force*

Base, Ohio, 1965), O. Shisha, Ed. (Academic Press, 1967), pp. 205–224.
40. D. Hilbert, Über die Darstellung definiter Formen als Summe von Formenquadraten. *Math. Ann.* **32**, 342–350 (1888).
41. K. G. Murty, S. N. Kabadi, Some NP-complete problems in quadratic and nonlinear programming. *Math. Program.* **39** (1987).
42. B. Reznick, Uniform denominators in Hilbert's seventeenth problem. *Math. Z.* **220** (1995).
43. A. W. Harrow, A. Natarajan, X. Wu, An improved semidefinite programming hierarchy for testing entanglement. *Commun. Math. Phys.* **352** (2017).
44. J. Britz, M. Laurent, Semidefinite hierarchies for diagonal unitary invariant bipartite quantum states. *arXiv:2512.06551 [quant-ph]* (2025).
45. K. Fang, H. Fawzi, The sum-of-squares hierarchy on the sphere and applications in quantum information theory. *Math. Program.* **190** (2021).
46. W. Hall, A new criterion for indecomposability of positive maps. *J. Phys. A: Math. Gen.* **39** (2006).
47. D. Chruściński, A. Kossakowski, How to construct indecomposable entanglement witnesses. *J. Phys. A: Math. Theor.* **41** (2008).
48. D. Chruściński, G. Sarbicki, Entanglement witnesses: construction, analysis and classification. *J. Phys. A: Math. Theor.* **47** (2014).
49. D. Chruściński, J. Pytel, G. Sarbicki, Constructing optimal entanglement witnesses. *Phys. Rev. A* **80** (2009).
50. H. Peyrl, P. A. Parrilo, Computing sum of squares decompositions with rational coefficients. *Theor. Comput. Sci.* **409** (2008).
51. K. Cafuta, I. Klep, J. Povh, Rational sums of hermitian squares of free noncommutative polynomials. *Ars Math. Contemp.* **9** (2015).
52. K. Życzkowski, K. A. Penson, I. Nechita, B. Collins, Generating random density matrices. *J. Math. Phys.* **52** (2011).
53. C. H. Bennett, *et al.*, Unextendible product bases and bound entanglement. *Phys. Rev. Lett.* **82** (1999).
54. E. Sindici, M. Piani, Simple class of bound entangled states based on the properties of the antisymmetric subspace. *Phys. Rev. A* **97** (2018).
55. J. Louvet, F. Damanet, T. Bastin, Bound entanglement in symmetric random induced states. *Phys. Rev. A* **113** (2026).
56. A. Gulati, I. Nechita, C. Pellegrini, Entanglement in the Dicke subspace. (2026). Available at: https://arxiv.org/abs/2602.15800.
57. L. M. Ioannou, B. C. Travaglione, Quantum separability and entanglement detection via entanglement-witness search and global optimization. *Phys. Rev. A* **73** (2006).
58. G. Chiribella, K. R. Davidson, V. I. Paulsen, M. Rahaman, Positive maps and entanglement in real Hilbert spaces. *Ann. Henri Poincaré* **24** (2023).
59. R. Augusiak, J. Grabowski, M. Kuś, M. Lewenstein, Searching for extremal PPT entangled states. *Opt. Commun.* **283** (2010).
60. P. Badziąg, K. Horodecki, M. Horodecki, J. Jenkinson, S. J. Szarek, Bound entangled states with extremal properties. *Phys. Rev. A* **90** (2014).

A. Example in 4 × 4

To make the pipeline concrete, we follow a single witness through the entire workflow, from its generation to both tests of the conjecture. Entry number 8172 of the generated library represents the strongest obtained detector. All numbers below are rounded for display; the exact rationalized certificate and the full-precision states are reproducible from the drivers in Section 2 under the fixed seed and archived in the repository under `code/results/`.

A.1. Generated witness. The KMŠZ construction (Section 1.4.5.1) returns a real positive but not completely positive map $\Phi : S_4 \to S_4$, equivalently the biquadratic form $p_\Phi(\boldsymbol{x}, \boldsymbol{y}) = \boldsymbol{y}^T \Phi(\boldsymbol{x}\boldsymbol{x}^T)\boldsymbol{y}$, a non-negative polynomial on $\mathbb{R}^4 \times \mathbb{R}^4$ that is not a sum of squares:

$$
\begin{aligned}
p_\Phi(\boldsymbol{x}, \boldsymbol{y}) = {} & 6.12x_4^2y_4^2 - 1.08x_4^2y_3y_4 + 2.51x_4^2y_3^2 - 0.28x_4^2y_2y_4 - 3.3x_4^2y_2y_3 + 2.08x_4^2y_2^2 + 0.07x_4^2y_1y_4 + 1.95x_4^2y_1y_3 \\
& -1.14x_4^2y_1y_2 + 0.84x_4^2y_1^2 - 1.9x_3x_4y_4^2 + 0.75x_3x_4y_3y_4 - 1.18x_3x_4y_3^2 - 5.84x_3x_4y_2y_4 + 0.28x_3x_4y_2y_3 \\
& -2.34x_3x_4y_2^2 - 2.05x_3x_4y_1y_4 - 1.07x_3x_4y_1y_3 - 0.12x_3x_4y_1y_2 - 2.23x_3x_4y_1^2 + 1.15x_3^2y_4^2 + 0.16x_3^2y_3y_4 \\
& +0.33x_3^2y_3^2 + 1.94x_3^2y_2y_4 + 0.75x_3^2y_2y_3 + 3.1x_3^2y_2^2 + 2.84x_3^2y_1y_4 + 0.14x_3^2y_1y_3 + 0.72x_3^2y_1y_2 + 2.19x_3^2y_1^2 \\
& -4.44x_2x_4y_4^2 + 0.13x_2x_4y_3y_4 + 1.14x_2x_4y_3^2 + 1.2x_2x_4y_2y_4 - 0.3x_2x_4y_2y_3 - 1.41x_2x_4y_2^2 - 6.72x_2x_4y_1y_4 \\
& -0.71x_2x_4y_1y_3 + 2.19x_2x_4y_1y_2 - 0.52x_2x_4y_1^2 + 2.71x_2x_3y_4^2 - 1.62x_2x_3y_3y_4 - 0.21x_2x_3y_3^2 + 4.83x_2x_3y_2y_4 \\
& +0.32x_2x_3y_2y_3 + 0.41x_2x_3y_2^2 + 2.96x_2x_3y_1y_4 - 2.6x_2x_3y_1y_3 + 4.05x_2x_3y_1y_2 - 0.5x_2x_3y_1^2 + 2.24x_2^2y_4^2 \\
& -2.04x_2^2y_3y_4 + 1.4x_2^2y_3^2 + 0.12x_2^2y_2y_4 - 1.26x_2^2y_2y_3 + 2.5x_2^2y_2^2 + 2.89x_2^2y_1y_4 - 0.9x_2^2y_1y_3 - 3.51x_2^2y_1y_2 \\
& +3.1x_2^2y_1^2 - 7.96x_1x_4y_4^2 + 2.36x_1x_4y_3y_4 - 4.68x_1x_4y_3^2 - 3.19x_1x_4y_2y_4 + 3.92x_1x_4y_2y_3 - 0.8x_1x_4y_2^2 \\
& -2.12x_1x_4y_1y_4 - 5.11x_1x_4y_1y_3 - 1.39x_1x_4y_1y_2 - 2.88x_1x_4y_1^2 - 0.05x_1x_3y_4^2 - 0.56x_1x_3y_3y_4 + 0.96x_1x_3y_3^2 \\
& +2.49x_1x_3y_2y_4 - 2.52x_1x_3y_2y_3 + 3.94x_1x_3y_2^2 + 2.9x_1x_3y_1y_4 + 3.25x_1x_3y_1y_3 + 5.52x_1x_3y_1y_2 + 3.93x_1x_3y_1^2 \\
& +1.95x_1x_2y_4^2 - 2.39x_1x_2y_3y_4 - 3.53x_1x_2y_3^2 - 2.42x_1x_2y_2y_4 + 2.53x_1x_2y_2y_3 + 1.2x_1x_2y_2^2 + 9.73x_1x_2y_1y_4 \\
& -0.55x_1x_2y_1y_3 + 3.79x_1x_2y_1y_2 - 0.75x_1x_2y_1^2 + 5.62x_1^2y_4^2 + 3.05x_1^2y_3y_4 + 4.56x_1^2y_3^2 + 2.53x_1^2y_2y_4 \\
& -3.92x_1^2y_2y_3 + 3.54x_1^2y_2^2 - 2.29x_1^2y_1y_4 - 0.11x_1^2y_1y_3 + 3.29x_1^2y_1y_2 + 5.9x_1^2y_1^2.
\end{aligned}
$$

This polynomial is what our construction actually produces. Collecting the product monomials into $\boldsymbol{v} = \boldsymbol{x} \otimes \boldsymbol{y} \in \mathbb{R}^{16}$ gives the quadratic representation $p_\Phi(\boldsymbol{x}, \boldsymbol{y}) = \boldsymbol{v}^T W \boldsymbol{v}$ of (17), from which we extract the entanglement witness $C_{\Gamma_\mathbb{C}}$ as the trivial complex extension (Section 1.4.5.2) of the gram matrix W. As the construction guarantees, W is symmetric, partial-transpose invariant (Section 1.4.5.2), block positive (Section 1.4.5.2), and not PSD, with eigenvalues running from -2.23 to 13.6 (four of them negative).

$$W \approx \begin{bmatrix}
5.9 & 1.64 & -0.06 & -1.15 & -0.38 & 0.95 & -0.14 & 2.43 & 1.97 & 1.38 & 0.81 & 0.73 & -1.44 & -0.35 & -1.28 & -0.53 \\
1.64 & 3.54 & -1.96 & 1.27 & 0.95 & 0.6 & 0.63 & -0.61 & 1.38 & 1.97 & -0.63 & 0.62 & -0.35 & -0.4 & 0.98 & -0.8 \\
-0.06 & -1.96 & 4.56 & 1.52 & -0.14 & 0.63 & -1.77 & -0.6 & 0.81 & -0.63 & 0.48 & -0.14 & -1.28 & 0.98 & -2.34 & 0.59 \\
-1.15 & 1.27 & 1.52 & 5.62 & 2.43 & -0.61 & -0.6 & 0.97 & 0.73 & 0.62 & -0.14 & -0.02 & -0.53 & -0.8 & 0.59 & -3.98 \\
-0.38 & 0.95 & -0.14 & 2.43 & 3.1 & -1.76 & -0.45 & 1.45 & -0.25 & 1.01 & -0.65 & 0.74 & -0.26 & 0.55 & -0.18 & -1.68 \\
0.95 & 0.6 & 0.63 & -0.61 & -1.76 & 2.5 & -0.63 & 0.06 & 1.01 & 0.2 & 0.08 & 1.21 & 0.55 & -0.71 & -0.08 & 0.3 \\
-0.14 & 0.63 & -1.77 & -0.6 & -0.45 & -0.63 & 1.4 & -1.02 & -0.65 & 0.08 & -0.11 & -0.41 & -0.18 & -0.08 & 0.57 & 0.03 \\
2.43 & -0.61 & -0.6 & 0.97 & 1.45 & 0.06 & -1.02 & 2.24 & 0.74 & 1.21 & -0.41 & 1.35 & -1.68 & 0.3 & 0.03 & -2.22 \\
1.97 & 1.38 & 0.81 & 0.73 & -0.25 & 1.01 & -0.65 & 0.74 & 2.19 & 0.36 & 0.07 & 1.42 & -1.11 & -0.03 & -0.27 & -0.51 \\
1.38 & 1.97 & -0.63 & 0.62 & 1.01 & 0.2 & 0.08 & 1.21 & 0.36 & 3.1 & 0.37 & 0.97 & -0.03 & -1.17 & 0.07 & -1.46 \\
0.81 & -0.63 & 0.48 & -0.14 & -0.65 & 0.08 & -0.11 & -0.41 & 0.07 & 0.37 & 0.33 & 0.08 & -0.27 & 0.07 & -0.59 & 0.19 \\
0.73 & 0.62 & -0.14 & -0.02 & 0.74 & 1.21 & -0.41 & 1.35 & 1.42 & 0.97 & 0.08 & 1.15 & -0.51 & -1.46 & 0.19 & -0.95 \\
-1.44 & -0.35 & -1.28 & -0.53 & -0.26 & 0.55 & -0.18 & -1.68 & -1.11 & -0.03 & -0.27 & -0.51 & 0.84 & -0.57 & 0.97 & 0.04 \\
-0.35 & -0.4 & 0.98 & -0.8 & 0.55 & -0.71 & -0.08 & 0.3 & -0.03 & -1.17 & 0.07 & -1.46 & -0.57 & 2.08 & -1.65 & -0.14 \\
-1.28 & 0.98 & -2.34 & 0.59 & -0.18 & -0.08 & 0.57 & 0.03 & -0.27 & 0.07 & -0.59 & 0.19 & 0.97 & -1.65 & 2.51 & -0.54 \\
-0.53 & -0.8 & 0.59 & -3.98 & -1.68 & 0.3 & 0.03 & -2.22 & -0.51 & -1.46 & 0.19 & -0.95 & 0.04 & -0.14 & -0.54 & 6.12
\end{bmatrix}$$

A.2. PPT entangled state. Minimising $\mathrm{tr}[W\rho]$ over the PPT cone via the SDP of (33) returns the state ρ, with $\mathrm{tr}[W\rho] = -3.48 \times 10^{-2} < 0$, certifying entanglement.

$$10^3 \cdot \rho \approx \begin{bmatrix}
24 & -17 & -3 & 11 & -4 & -2 & 2 & -15 & 14 & 4 & 10 & -3 & 38 & 32 & 25 & 0 \\
-17 & 81 & 46 & -48 & -2 & -55 & -41 & 37 & 4 & -18 & -11 & 5 & 32 & -14 & -1 & 6 \\
-3 & 46 & 73 & -55 & 2 & -41 & 3 & 26 & 10 & -11 & 23 & -17 & 25 & -1 & 35 & -27 \\
11 & -48 & -55 & 59 & -15 & 37 & 26 & -4 & -3 & 5 & -17 & 14 & 0 & 6 & -27 & 24 \\
-4 & -2 & 2 & -15 & 64 & 13 & -2 & -46 & 45 & -1 & 10 & -25 & 12 & -44 & -9 & -5 \\
-2 & -55 & -41 & 37 & 13 & 52 & 35 & -22 & -1 & 13 & -2 & -11 & -44 & -16 & -21 & 2 \\
2 & -41 & 3 & 26 & -2 & 35 & 79 & 19 & 10 & -2 & 9 & -16 & -9 & -21 & -12 & -9 \\
-15 & 37 & 26 & -4 & -46 & -22 & 19 & 79 & -25 & -11 & -16 & 3 & -5 & 2 & -9 & 10 \\
14 & 4 & 10 & -3 & 45 & -1 & 10 & -25 & 59 & -6 & 8 & -25 & 64 & -20 & 3 & 6 \\
4 & -18 & -11 & 5 & -1 & 13 & -2 & -11 & -6 & 13 & 2 & -12 & -20 & 17 & 12 & -2 \\
10 & -11 & 23 & -17 & 10 & -2 & 9 & -16 & 8 & 2 & 30 & -15 & 3 & 12 & 32 & -25 \\
-3 & 5 & -17 & 14 & -25 & -11 & -16 & 3 & -25 & -12 & -15 & 51 & 6 & -2 & -25 & 9 \\
38 & 32 & 25 & 0 & 12 & -44 & -9 & -5 & 64 & -20 & 3 & 6 & 149 & 19 & 22 & 20 \\
32 & -14 & -1 & 6 & -44 & -16 & -21 & 2 & -20 & 17 & 12 & -2 & 19 & 90 & 60 & -3 \\
25 & -1 & 35 & -27 & -9 & -21 & -12 & -9 & 3 & 12 & 32 & -25 & 22 & 60 & 69 & -24 \\
0 & 6 & -27 & 24 & -5 & 2 & -9 & 10 & 6 & -2 & -25 & 9 & 20 & -3 & -24 & 29
\end{bmatrix}$$

Due to the imposed constraints, ρ is PPT: both ρ and ρ^Γ are PSD to numerical tolerance (smallest eigenvalues $\approx 9 \times 10^{-14}$ and $\approx -4 \times 10^{-12}$).

A.3. Composition of two PPT maps. We minimize $\mathrm{tr}[WC_{\Phi\circ\Psi}]$ over compositions of two PPT channels via the see-saw SDP of (34), to obtain

$$10^3 \cdot C_{\Phi\circ\Psi} \approx \begin{bmatrix}
26 & -33 & -17 & 20 & 9 & 19 & 5 & -38 & 2 & -8 & -42 & 14 & -4 & 3 & 15 & -3 \\
-33 & 66 & 48 & -64 & 19 & -34 & -29 & 31 & -8 & 21 & 17 & -23 & 3 & -5 & -5 & 5 \\
-17 & 48 & 80 & -59 & 5 & -29 & 31 & 33 & -42 & 17 & -38 & 28 & 15 & -5 & 11 & -10 \\
20 & -64 & -59 & 77 & -38 & 31 & 33 & -6 & 14 & -23 & 28 & 15 & -3 & 5 & -10 & -2 \\
9 & 19 & 5 & -38 & 49 & -3 & -54 & -45 & 12 & 10 & -44 & -23 & -5 & -1 & 13 & 5 \\
19 & -34 & -29 & 31 & -3 & 18 & 4 & -24 & 10 & -11 & -11 & 6 & -1 & 2 & 2 & -1 \\
5 & -29 & 31 & 33 & -54 & 4 & 102 & 44 & -44 & -11 & -12 & 68 & 13 & 2 & 5 & -17 \\
-38 & 31 & 33 & -6 & -45 & -24 & 44 & 80 & -23 & 6 & 68 & 10 & 5 & -1 & -17 & -2 \\
2 & -8 & -42 & 14 & 12 & 10 & -44 & -23 & 42 & -8 & 18 & -35 & 29 & -15 & -37 & 6 \\
-8 & 21 & 17 & -23 & 10 & -11 & -11 & 6 & -8 & 9 & 2 & -6 & -15 & 5 & 13 & 3 \\
-42 & 17 & -38 & 28 & -44 & -11 & -12 & 68 & 18 & 2 & 145 & -41 & -37 & 13 & -20 & 17 \\
14 & -23 & 28 & 15 & -23 & 6 & 68 & 10 & -35 & -6 & -41 & 53 & 6 & 3 & 17 & -14 \\
-4 & 3 & 15 & -3 & -5 & -1 & 13 & 5 & 29 & -15 & -37 & 6 & 136 & -58 & -95 & -17 \\
3 & -5 & -5 & 5 & -1 & 2 & 2 & -1 & -15 & 5 & 13 & 3 & -58 & 25 & 42 & 6 \\
15 & -5 & 11 & -10 & 13 & 2 & 5 & -17 & -37 & 13 & -20 & 17 & -95 & 42 & 83 & 6 \\
-3 & 5 & -10 & -2 & 5 & -1 & -17 & -2 & 6 & 3 & 17 & -14 & -17 & 6 & 6 & 5
\end{bmatrix}$$

with minimum eigenvalues of $C_{\Phi\circ\Psi}$ and $C^\Gamma_{\Phi\circ\Psi}$ both $\approx 4 \times 10^{-14}$. Neither our witness nor the level-2 DPS relaxation detect entanglement for this state, bottoming out at $\mathrm{tr}[WC_{\Phi\circ\Psi}] = 5.5 \times 10^{-2} \geq 0$.